

2026 FALL SEMESTER
Programming for Data Science Lab (MACSE502) Record
Practice Program

Date:17/07/26

Week No.: 2

Program S. No. 2

Title	Data Manipulation and Analysis using the Starwars Dataset in R																																	
Problem Statement	Data Cleaning and Preprocessing with Pandas: Create a DataFrame with some missing values in columns Name, Age, and City. Perform the following operations: fill missing values in the Age column with the mean age, drop rows where Name or City is missing, and convert the Age column to integer type. Finally, normalize the Age column using min-max scaling.																																	
Objectives	● To understand how to create a Pandas DataFrame containing missing values. ● To learn how to identify and handle missing data using the fillna() function. ● To remove incomplete records using the dropna() function. ● To convert a DataFrame column to the required data type using the astype() function. ● To normalize numerical data using the Min-Max Scaling technique. ● To understand the importance of data preprocessing before performing data analysis and machine learning tasks.																																	
Dataset URL Link / File	<table><tr><th>Name</th><th>Age</th><th>City</th></tr><tr><td>Hasnain Makada</td><td>20.0</td><td>Delhi</td></tr><tr><td>Tim</td><td>np.nan</td><td>Mumbai</td></tr><tr><td>np.nan</td><td>25.0</td><td>Pune</td></tr><tr><td>Cook</td><td>30.0</td><td>np.nan</td></tr><tr><td>Nolan</td><td>np.nan</td><td>Chennai</td></tr><tr><td>Heisenberg</td><td>35.0</td><td>Delhi</td></tr><tr><td>True Detective</td><td>40.0</td><td>Jaipur</td></tr><tr><td>Casio</td><td>45.0</td><td>Surat</td></tr><tr><td>MSI</td><td>50.0</td><td>np.nan</td></tr></table>			Name	Age	City	Hasnain Makada	20.0	Delhi	Tim	np.nan	Mumbai	np.nan	25.0	Pune	Cook	30.0	np.nan	Nolan	np.nan	Chennai	Heisenberg	35.0	Delhi	True Detective	40.0	Jaipur	Casio	45.0	Surat	MSI	50.0	np.nan	
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Functions Details	Sl. No.	Function	Package	Description of function																														
	1	DataFrame()	pandas	Creates a DataFrame from structured data																														
	2	mean()	pandas	Calculates the average value																														
	3	fillna()	pandas	Replaces missing values with a specified value.																														
	4	dropna()	pandas	Removes rows containing missing values.																														

Student Reg. No: 26MML0031

Name: Hasnain Makada

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore

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	5	astype()	pandas	Converts a column to the specified data type.
	6	min()	pandas	Returns the minimum value of a column.
	7	max()	pandas	Returns the maximum value of a column.
	8	np.nan	pandas	Represents missing (null) values in the dataset.
Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	Step 1 Import the required libraries, Pandas and NumPy, to perform data manipulation and handle missing values. Step 2 Create a dictionary containing the Name, Age, and City columns, with some missing values represented using np.nan. Step 3 Convert the dictionary into a Pandas DataFrame using the DataFrame() function and display the original dataset. Step 4 Calculate the mean value of the Age column using the mean() function and replace all missing age values using the fillna() function. Step 5 Remove all rows where either the Name or City column contains missing values using the dropna() function. Step 6 Convert the Age column from floating-point type to integer type using the astype() function. Step 7 Normalize the Age column using the Min-Max Scaling formula: $\text{Normalized Age} = \frac{\text{Age} - \text{Minimum Age}}{\text{Maximum Age} - \text{Minimum Age}}$ Step 8 Display the cleaned and normalized DataFrame.			
Program	<pre># Import the required libraries import pandas as pd import numpy as np # Step 1: Create a personalized dataset with missing values student_data = [["Hasnain Makada", 20, "Delhi"], ["Tim", np.nan, "Mumbai"], [np.nan, 25, "Pune"], ["Cook", 30, np.nan], ["Nolan", np.nan, "Chennai"],</pre>			

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```
["Heisenberg", 35, "Delhi"],
["True Detective", 40, "Jaipur"],
["Casio", 45, "Surat"],
["MSI", 50, np.nan]
]

# Step 2: Convert the dataset into a DataFrame
student_df = pd.DataFrame(student_data, columns=["Name", "Age",
"City"])

# Display the original DataFrame
print("Original DataFrame")
print(student_df)

# Step 3: Calculate the mean age
mean_age = student_df["Age"].mean()

# Fill missing values in the Age column with the mean age
student_df["Age"] = student_df["Age"].fillna(mean_age)
print("\nAfter filling missing Age values")
print(student_df)

# Step 4: Drop rows where Name or City is missing
student_df = student_df.dropna(subset=["Name", "City"]).copy()
print("\nAfter dropping rows with missing Name or City")
print(student_df)

# Step 5: Convert the Age column to integer type
student_df["Age"] = student_df["Age"].astype(int)
print("\nAfter converting Age to integer")
print(student_df)

# Step 6: Normalize the Age column using Min-Max Scaling
min_age = student_df["Age"].min()
max_age = student_df["Age"].max()

student_df["Normalized_Age"] = (
    (student_df["Age"] - min_age) / (max_age - min_age)
)

# Display the final DataFrame
print("\nFinal cleaned and normalized DataFrame")
print(student_df)
```

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**Output
Snapshots
With
Explanation/
Observations**

```
Original DataFrame
|   |   |   | Name | Age | City |
|---|---|---|-----|-----|-----|
0  | Hasnain | Makada | 20.0 | Delhi |
1  |         | Tim    | NaN  | Mumbai |
2  |         | NaN    | 25.0 | Pune |
3  |         | Cook   | 30.0 | NaN |
4  |         | Nolan  | NaN  | Chennai |
5  |         | Heisenberg | 35.0 | Delhi |
6  | True Detective | 40.0 | Jaipur |
7  |         | Casio  | 45.0 | Surat |
8  |         | MSI    | 50.0 | NaN |
```

Figure 1.1 Original Data Frame

Figure 1.1 Original DataFrame

Explanation:

Figure 1.1 shows the original DataFrame created from the employee dataset. The DataFrame contains three columns: **Name**, **Age**, and **City**. Several records contain missing values (NaN) in

different columns. Such incomplete datasets are common in real-world applications and require preprocessing before analysis. This DataFrame serves as the input for all subsequent data cleaning operations.

Figure 1.2 DataFrame after Filling Missing Age Values

Explanation:

Figure 1.2 displays the DataFrame after replacing the missing values in the **Age** column with the average (mean) age using the fillna() function. The missing numerical values have been successfully filled while the missing values in the **Name** and **City** columns remain unchanged. This technique helps preserve valuable records instead of removing them completely.

```
After filling missing Age values
|   |   |   | Name | Age | City |
|---|---|---|-----|-----|-----|
0  | Hasnain | Makada | 20.0 | Delhi |
1  |         | Tim    | 35.0 | Mumbai |
2  |         | NaN    | 25.0 | Pune |
3  |         | Cook   | 30.0 | NaN |
4  |         | Nolan  | 35.0 | Chennai |
5  |         | Heisenberg | 35.0 | Delhi |
6  | True Detective | 40.0 | Jaipur |
7  |         | Casio  | 45.0 | Surat |
8  |         | MSI    | 50.0 | NaN |
```

Figure 1.2 DataFrame with mean age

```
After dropping rows with missing Name or City
|   |   |   | Name | Age | City |
|---|---|---|-----|-----|-----|
0  | Hasnain | Makada | 20.0 | Delhi |
1  |         | Tim    | 35.0 | Mumbai |
4  |         | Nolan  | 35.0 | Chennai |
5  |         | Heisenberg | 35.0 | Delhi |
6  | True Detective | 40.0 | Jaipur |
7  |         | Casio  | 45.0 | Surat |
```

Figure 1.3 DataFrame with no missing rows

Figure 1.3 DataFrame after Removing Missing Name and City Values

Explanation:

Figure 1.3 presents the cleaned DataFrame after removing all rows containing missing values in either the **Name** or **City** columns using the dropna() function. Only complete records remain in the dataset, making it suitable for further processing and analysis. Removing incomplete records improves data quality and reduces inconsistencies.

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Figure 1.4 DataFrame after Converting Age to Integer

Explanation:

Figure 1.4 illustrates the DataFrame after converting the **Age** column from floating-point values to integer values using the `astype()` function. Since the

missing values were replaced with the mean, the column was automatically converted to float. Converting it back to integer improves data consistency and makes the dataset more suitable for applications where age is represented as whole numbers.

```
After converting Age to integer
|  |  |  | Name | Age | City |
| 0 | Hasnain | Makada | 20 | Delhi |
| 1 |  | Tim | 35 | Mumbai |
| 4 |  | Nolan | 35 | Chennai |
| 5 |  | Heisenberg | 35 | Delhi |
| 6 | True Detective | 40 | Jaipur |
| 7 |  | Casio | 45 | Surat |
```

Figure 1.4 DataFrame with age col as int

```
Final cleaned and normalized DataFrame
|  |  |  | Name | Age | City | Normalized_Age |
| 0 | Hasnain | Makada | 20 | Delhi | 0.0 |
| 1 |  | Tim | 35 | Mumbai | 0.6 |
| 4 |  | Nolan | 35 | Chennai | 0.6 |
| 5 |  | Heisenberg | 35 | Delhi | 0.6 |
| 6 | True Detective | 40 | Jaipur | 0.8 |
| 7 |  | Casio | 45 | Surat | 1.0 |
```

Figure 1.5 DataFrame with normalized age

Figure 1.5 DataFrame after Min-Max Normalization

Explanation:

Figure 1.5 shows the final DataFrame after applying **Min-Max Scaling** to the **Age** column. A new column named **Normalized_Age** has been added, where all age values are scaled between **0** and **1**. The smallest age receives a value of **0**, while the largest age receives a value of **1**. Feature normalization is an important preprocessing technique that ensures all numerical attributes have a comparable scale, improving the performance of many machine learning algorithms.

Conclusions

This experiment successfully demonstrated the process of **data cleaning and preprocessing** using the Pandas library. A DataFrame containing missing values in the **Name**, **Age**, and **City** columns was created and processed using various data preprocessing techniques.

The following operations were successfully performed:

- Missing values in the **Age** column were replaced with the mean age using the `fillna()` function.
- Rows containing missing values in the **Name** or **City** columns were removed using the `dropna()` function.
- The **Age** column was converted from floating-point to integer data type using the `astype()` function.
- The **Age** values were normalized using **Min-Max Scaling**, bringing all values into the range of **0** to **1**.

Overall, this experiment provided practical knowledge of handling missing data, cleaning datasets, performing data type conversion, and applying feature scaling techniques. These preprocessing methods are essential steps in **Data**

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	Science, Machine Learning, Data Analytics, and Artificial Intelligence, as they improve data quality and prepare datasets for accurate analysis and predictive modeling.
References	1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data with R, Addison-Wesley, 2018 2. Python Program https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/26MML0031_WEEK2_PRACTICE2.py 3. Class lecture link https://drive.google.com/file/d/15diEukCtrPLimox63oa9ce1Ut4qVMUy/view?usp=drive_link